Hilbert early stop on the Joux���Vitse Nagao arm at scale

Hilbert early stop on the Joux–Vitse Nagao arm at p = 1301–1601 (2026-11-12)

Lead: `LEAD-C-jv-nagao-hilbert-at-scale`, stage 1 (counts) and stage 2 (rescore), preregistered in `research/leads_20261111/lane_C_prime_ext_cover.md` before any of these runs. Run because the owner asked for it.

All results are `[pcc:unverified]`: - local L0 runs on a 4-vCPU cloud container; - no replay certificate, no validator report; - the rho reference is the pooled one from another experiment.

The change is classified as **engineering**: the counted cost falls while the emitted native counter stays flat. It is not an advance.

What was run

- **Binary.** `target/release/examples/jv_cover`, built from the commit in `results/SOURCE_COMMIT`, sha256 in `results/BINARY.sha256`.
- **Arm A (baseline).** The frozen line-search records `research/jv_cover_fbline_20261102/results/line_p{p}_s{s}.json`.
- **Drift control.** `aa_line_p1301_s1` re-runs the frozen command on today's binary. It reproduces the frozen record exactly: `total_muls` 265,597,174,434, `f4_muls`, `f4_native_ops`, `relation_digest`, relations, residuals and S are all identical.
- **Arm B.** The same 9 instances (p ∈ {1301, 1451, 1601}, seeds 1–3), the same command, plus `--stop hilbert:16`. Driver: `run.sh`.
- **Stage 1.** `compare.py`, output in `results/paired.txt` and `results/paired.json`.
- **Stage 2.** The unchanged `negrho_rescore` tool from `research/jv_cover_negrho_rescore_20261101/fit` was run on a copy of that study's results directory, with `inputs/cover/dlp_p*` replaced by the arm-B records. This is the procedure the line-search study used. Output is in `rescore/`.
  - The tool prints "consistency controls FAILED". That is expected. Those controls check that the *old* cover records reproduce the old study's numbers, and they were replaced on purpose.
  - The rho reference itself reproduces: steps/√l 0.8851 ± 0.0537, against the reported 0.885 ± 0.054.

Stage 1: counts, paired

p	seeds	S_B / S_A	f4_muls ratio	f4_native_ops ratio	relations / digest	Hilbert fallbacks
1301	1, 2, 3	0.5278, 0.5276, 0.5279	0.3895	0.9830	identical 3/3	227, 269, 219
1451	1, 2, 3	0.5280, 0.5278, 0.5280	0.3895	0.9830	identical 3/3	270, 253, 266
1601	1, 2, 3	0.5277, 0.5278, 0.5277	0.3894	0.983	identical 3/3	244, 249, 222

The fallbacks are out of 265,687–368,577 systems with the stop applied, about 0.08 %.

The preregistered falsifiers all pass:

Falsifier	Result
F1: any digest, relation or logarithm mismatch	none, 9 of 9 identical
F2: upper 95 % bound of S_B/S_A above 0.70	all ratios 0.5276–0.5280
F3: fallbacks above 25 %	about 0.08 %
F4: native counter up more than 1.10×	0.983×

The prediction was 0.54 [0.50, 0.60]. The measured ratio is a 1.895× reduction in counted multiplications at equal relations.

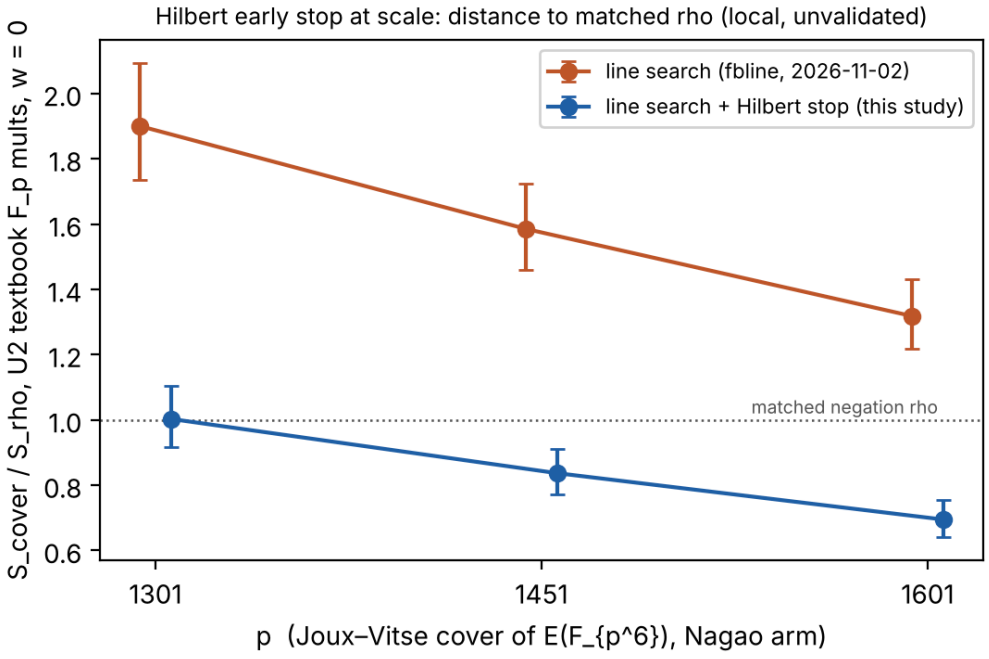
Stage 2: distance to matched negation rho

Pooled reference (b), 126 runs, `rescore/table.md`:

unit	p = 1301	p = 1451	p = 1601
U2 textbook F_p mults, w = 0, before (line search)	1.900 above	1.585 above	1.318 above
U2, w = 0, with Hilbert stop	**1.003 [0.916, 1.104]**, not resolved	**0.837 [0.771, 0.910]**, below	**0.695 [0.642, 0.756]**, below
U2, w = 0.1 / 1	0.861 / 0.831, below	0.718 / 0.693, below	0.597 / 0.576, below
order holds across w ∈ {0, 0.1, 1}	**no**	**yes (below)**	**yes (below)**
U1 group additions, w = 0	0.417 below	0.348 below	0.289 below
modelled U2 crossover p*	1327 [1278, 1381]		

The prediction was 1.03 / 0.86 / 0.71. Figure: `u2_before_after.{svg,png,pdf}`.

Reading. On these instances, and in this accounting, the Nagao arm of the Joux–Vitse cover moves from 1.3–1.9× above matched negation rho in U2 to below it at p = 1451 and 1601. There the order holds at all three §5a weights. At p = 1301 it is level with rho (not resolved) `[pcc:unverified]`.



What this is not

- **Not a leaderboard or benchmark row.**
  - The validation gate needs a replay certificate, a validator report and a matched same-binary rho reference.
  - The benchmark-record rule needs raw data in S3.
  - Neither is possible from this session, because its AWS identity is rejected. The lead's stage-3 Iso Lab job (9 instances × 3 arms, emitted F4-driver counters, 16 fresh rho runs per prime) is the route to a gated row.
- **The cover count is a lower bound.** F_p additions are not emitted on the cover side (`rescore/table.md`, unpriced-counter note). A "below" verdict could narrow under complete counts; the w = 0.1 and w = 1 columns bound this.
- **Not a statement about any deployed curve.** It concerns the weak class of curves over F_{p^6} with a Joux–Vitse cover, at p ≤ 1601.